

Youfu Yan

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SUMMARY

Software Development Engineer with 2+ years at Amazon building and operating production AI/ML and distributed systems across AWS. Owns ambiguous cross-team work from architecture and model evaluation through implementation, staged rollout, and on-call operations. Equal-contribution first author of IEEE VIS 2024 Honorable Mention research on LLM and Knowledge Graph systems.

TECHNICAL SKILLS

AI/ML & GenAI: Amazon SageMaker, AWS Bedrock, PyTorch, LLM-as-a-Judge evaluation, pairwise A/B testing, RAG, Knowledge Graphs, prompt engineering, image matting/segmentation

Languages: Java, Python, TypeScript, JavaScript, SQL, C/C++, Bash

Backend & Cloud: Spring Boot, REST APIs, AWS Lambda, DynamoDB, SQS/SNS, S3, CloudWatch, event-driven architecture, microservices, OAuth 2.0, multi-region deployment, AWS CDK, Docker

Frontend & Quality: React, Redux, Next.js, React Native, D3.js, Playwright, Cypress, JUnit, Mockito, Jest, integration/load/stress testing

EXPERIENCE

Amazon - Software Development Engineer (June 2024 - Present | Seattle, WA)

AI/ML Platform Migration | SageMaker, Bedrock, Python, PyTorch (2026)

Designed and drove an AWS-native background-removal migration plan for approximately **1M uncached image requests per month**; projected to reduce annual run-rate by about **\$96K (approximately 80%)** versus the third-party API, based on current traffic, \$0.01/request vendor pricing, and an estimated \$2K/month hosted endpoint.

Built a **GPU SageMaker real-time endpoint** and a Python benchmark harness that compared five model and service backends in parallel across output quality, client latency, and server-side model latency.

Designed a blinded **LLM-as-a-judge** pipeline using three-vote majority, five matting criteria, and pairwise position-swap controls over 500 production images; identified and corrected an input-representation issue that had produced approximately **70-79% false-failure verdicts**.

Distributed Systems & Production Reliability | Java, AWS, DynamoDB, React (2024-2026)

Owned the launch of a business-critical content API in a new AWS region and executed a staged **1% to 100% traffic shift**; diagnosed a live certificate mismatch at 20%, rolled back safely, coordinated the fix, and completed the launch with **no customer-facing outage**.

Designed and built the first version of an event-driven creator-notification service used across three teams, including **event-time DynamoDB conditional writes** and retry/DLQ handling to preserve correctness under out-of-order events.

Drove production content-ingestion workflows for a program supporting approximately **2,300 creators and 85K ingested items**; contributed to a scheduling platform serving **8,000+ creators across 256K+ workflows**.

Improved operational readiness through on-call incident response, runbooks, load and stress testing, a production React race-condition investigation, and migration of a cross-origin E2E suite from Cypress to Playwright.

Amazon - Software Development Engineer Intern (June 2023 - August 2023 | New York, NY)

Built a cross-platform real-time notification system using Java, React Native, AWS Lambda, and SQS, reducing content-status notification latency **60%** (150 ms to 60 ms) across iOS and Android.

Partnered with UX, QA, frontend, and backend teams to shorten the release cycle from four weeks to three.

University of Minnesota, Twin Cities - Research Assistant, AI/ML & Human-Centered Computing Lab (September 2023 - May 2024 | Minneapolis, MN)

Co-developed KNOWNet, a conversational health-information system combining OpenAI GPT-4, RAG, Neo4j Knowledge Graphs, Flask/Next.js, and progressive D3.js visualization; the reported evaluation showed a **20% improvement in user comprehension**.

PUBLICATION

KNOWNET: Guided Health Information Seeking from LLMs via Knowledge Graph Integration (2024)

Youfu Yan*, Yu Hou*, Yongkang Xiao, Rui Zhang, Qianwen Wang (*equal contribution) | *IEEE TVCG* | **IEEE VIS 2024 Honorable Mention**

EDUCATION

University of Minnesota, Twin Cities	M.S. in Computer Science, GPA: 3.95 TA: Data Structures & Algorithms	2024
University of Minnesota, Twin Cities	M.A. in Quantitative Methods in Education	2022
University of California, Santa Barbara	M.A. in Statistics	2020
Certification: AWS Certified Solutions Architect		